Andrés Sánchez Marín

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PROFESSIONAL EXPERIENCE

Amazon — Madrid, Spain
Senior Software Security Engineer — **Sep 2024 – present**
Device OS Security

HexHive group, EPFL — Lausanne, Switzerland
Research Assistant — **Sep 2019 – Aug 2024**
Compartmentalization techniques evaluation, compiler-assisted language compartmentalization. Automating the detection of Speculative ROP chains. Software Fault Isolation of legacy code through WebAssembly. Compiler-assisted hardening of SGX controlled channel attacks. Teaching assistant of: CS-412 Software Security (2022), COM-402 Information Security and Privacy (2022, 2023), CS-420 Advanced Compiler Construction (2023, 2024)

IBM Research — Zürich, Switzerland
Research Assistant — **Jun 2020 – Sep 2020**
Taxonomy of kernel code reuse attacks, KASLR bypasses and mitigations in software and architectural level.

Massachusetts Institute of Technology — Cambridge, United States
Visiting Researcher — **Jun 2019 – Aug 2019**
Security analysis of compressed cache architectures, demonstration of first data at rest microarchitectural leak.

IMDEA Software Institute — Madrid, Spain
Research Intern — **Sep 2018 – Jun 2019**
Evaluating countermeasures for speculative execution leaks, automating their detection in large code-bases.

Distributed Systems Laboratory, UPM — Madrid, Spain
Research Intern — **Sep 2017 – Dec 2017**
Integrating an Elastic Complex Event Processing for Static and Streaming Data in a NoSQL distributed database.

EDUCATION

École Polytechnique Fédrale de Lausanne — Lausanne, Switzerland
Master of Science in Computer Science — **2019 – 2021**
MSc Thesis: Characterization of the overheads for comprehensive compartmentalization of software

Universidad Politécnica de Madrid — Madrid, Spain
Bachelor of Science in Mathematics and Computing — **2015 – 2019**
Member of ACM UPM chapter **BSc Thesis:** Detecting speculative information flows on large code-bases

AREAS OF INTEREST

Computer Systems	Security	Compilers
Hardware Architecture	Operating Systems	Formal Verification

Honors & Awards	Distinquished Artifact Evaluator Award — Eurosys'23 IEEE Micro Top Picks Award — 2021; MSc Research Scholar, EPFL — 2019 – 2021 Obtained honors in 10 courses during my BSc studies
Service	AE Eurosys'23, AE ISCA'23, AE NDSS'24
Languages	**Spanish** (mother tongue), **English** (advanced), **French** (intermediate)
Programming	C, Rust, C++, ASM, WASM, Python, Linux, Git, LLVM
Certifications	CCNA 1 & 2, Free Time Monitor Certification

PROCEEDINGS PUBLICATIONS

[1] M. Guarnieri, B. Koepf, J. F. Morales, J. Reineke, and A. Sánchez, "SPECTECTOR: Principled detection of speculative information flows," in *Proceedings of the 41st IEEE Symposium on Security and Privacy*, IEEE, 2020.

[2] P.-A. Tsai, A. Sanchez, C. Fletcher, and D. Sanchez, "Safecracker: Leaking secrets through compressed caches," in *Proceedings of the Twenty-Fifth International Conference on Architectural Support for Programming Languages and Operating Systems*, ASPLOS '20, ACM, 2020.

[3] A. Bhattacharyya, F. Hofhammer, Y. Li, S. Gupta, A. Sanchez, B. Falsafi, and M. Payer, "Securecells: A secure compartmentalized architecture," in *2023 IEEE Symposium on Security and Privacy (SP)*, pp. 2921–2939, IEEE, 2023.

[4] D. Vanoverloop, A. Sánchez, F. Toffalini, F. Piessens, M. Payer, and J. V. Bulck, "TLBlur: Compiler-Assisted automated hardening against controlled channels on Off-the-Shelf intel SGX platforms," in *34th USENIX Security Symposium (USENIX Security 25)*, (Seattle, WA), pp. 1167–1186, USENIX Association, Aug. 2025.

[5] A. Bhattacharyya, A. Sánchez, E. M. Koruyeh, N. Abu-Ghazaleh, C. Song, and M. Payer, "Specrop: Speculative exploitation of ROP chains," in *23rd International Symposium on Research in Attacks, Intrusions and Defenses (RAID 2020)*, (San Sebastian), pp. 1–16, USENIX Association, Oct. 2020.

SUPERVISED STUDENT PROJECTS

Vicenzo Pellegrini — Pitfalls and characterization of C libraries isolation through WebAssembly, Fall'23
Andrija Jelenkovic — Characterization of interface definition bugs for Rust projects, Fall'23
Lachat Niels Marco — Exploring and evaluating the safety of internal Android language interfaces, Spring'23
Edouard Michelin — A study on the overhead of memory-tagging in compression libraries, Spring'23
Francesco Berla — A study of WebAssembly-based SFI through safe intermediate representations, Fall'22
Maxime Würsch — OpenSSL keys isolation benchmark suite, Spring'22
Basil Ottinger — Unikernel adoption of novel compartmentalization mechanisms, Spring'22
Fréderic Julien — FreeBSD kernel adoption of novel MPK techniques, Fall'21